

World Models as Intermediaries for Scaling Agents in High-Cost Environments

Overview

Large Language Model (LLM) agents have achieved superhuman performance in low-cost digital environments like coding and competitive mathematics. However, these successes have plateaued in complex, high-stakes domains such as robotics, machine learning (ML) engineering, and scientific discovery. The primary bottleneck is the prohibitively high cost of real-world interaction—measured in execution time, physical resources, and safety risks. This proposal seeks to bridge this gap by developing a unified framework that utilizes generalist world models as an intermediary between agents and the real world. By compressing real-world data into learned simulators that model dynamics, rewards, execution costs, and task distributions, we transform expensive, sequential physical interactions into low-cost, parallelizable digital queries. This paradigm shifts agent learning from brittle hand-coded simulators or slow physical trials to high-fidelity, data-driven "sandboxes," enabling safe, sample-efficient reinforcement learning (RL) before real-world deployment.

Keywords: World Models; Reinforcement Learning; Autonomous Agents; Model-Based Learning; High-Cost Environments; Scientific Discovery

Intellectual Merit

This research develops a comprehensive framework for training agents via learned world models through four integrated research thrusts: **(1) Large-Scale Dataset Curation and Passive Logging:** To overcome the scarcity of action-rich data in physical and specialized domains, this thrust will synthesize control data from internet-scale sources. We will leverage image/video captioning and inverse dynamics models to pseudo-label vast quantities of action-free videos. Concurrently, we will develop passive logging mechanisms to automatically capture control inputs, system observations, and experimental outcomes from high-cost workflows like ML engineering pipelines and automated laboratory equipment. **(2) Unified Architectures for Dynamics, Rewards, and Task Distributions:** We will develop scalable Transformer-based architectures to instantiate a comprehensive world model comprising three core components: dynamics, rewards/costs, and task distributions. This involves designing specific encoders for structured data, 3D geometry, and video to accurately predict future states and execution costs. By capturing realistic task distributions directly from interaction logs, the world model will act not just as a physics simulator, but as an automated curriculum generator. **(3) Time-Compressed Planning and Iterative RL Optimization:** This thrust leverages the world model for low-cost, rapid policy evaluation and RL optimization. By simulating long-horizon dynamics—such as molecular evolution or application UI state changes—in a time-compressed manner, agents can perform rigorous planning and off-policy learning. This avoids the severe gradient variance and extreme off-policy instability that typically plague asynchronous RL in environments with delayed real-world feedback. **(4) Modular Evaluation and Continuous Real-World Calibration:** We will design a closed-loop evaluation protocol based on the classical Dyna architecture. Agents will be primarily trained in the world model and occasionally deployed in the real world to validate end-to-end performance. The resulting physical interaction data will not only verify the policy but will be recycled to iteratively refine the world model's physical and functional validity, closing the sim-to-real gap.